

Insurance Portfolio Review

SQL validation and simple Python analysis

This report adds a practical analysis layer to the existing Power BI dashboard. The SQL work checks the main dashboard numbers and the Python work asks additional questions about portfolio mix, premium, customers, agents, and data quality.

What was delivered

The analysis uses PostgreSQL queries for KPI validation and additional business questions. The Python file uses Pandas, NumPy, Matplotlib, and Seaborn with simple grouped summaries, percentages, ratios, and correlations.

Main takeaways

Active policies account for most records and annual premium. Premium is concentrated in a small number of states and plans. Modeled paid premium is much smaller than modeled lifetime premium, and repeated policy numbers mean that row counts and distinct-policy counts should be kept separate.

SQL: Dashboard KPI Validation

Purpose: These PostgreSQL checks recalculate the main numbers shown in the Power BI dashboard. They help confirm that the report agrees with the policy fact table after data loading or refresh.

What is validated

- Portfolio size: total rows and distinct policy numbers.
- Portfolio status: active records and active percentage.
- Premium measures: annual premium, lifetime premium, modeled paid premium, and paid share.
- Concentration measures: the two largest protection plans and the three largest states by active premium.

How the SQL works

The queries use COUNT and COUNT DISTINCT for record checks, CASE expressions to convert monthly and quarterly payments into annual premium, SUM for portfolio totals, and ranking functions to identify the largest plans and states. The paid-premium query uses the report snapshot date and limits paid years to the stated policy tenure.

Why it matters

Running these checks gives the dashboard a clear control layer. If a number does not match, the analyst can investigate filters, duplicate policy numbers, premium annualization, date logic, or the active-status definition before the KPI is shared with management.

Important: modeled paid premium is an analytical estimate. It should not be described as verified cash collections or as an overdue balance.

SQL: Additional Business Insights

Purpose: These ten questions go beyond the dashboard headline KPIs and identify areas for management review.

What is explored

- Status risk by state and purchase-year cohort.
- Annual premium per record by protection plan and policy type.
- Coverage differences by payment frequency.
- Premium differences by smoker status.
- Agent lapse rates and annual premium per policy.
- Region mapping completeness and basic premium, date, tenure, and claim-status warnings.

How to use the results

The queries help management choose where to look next. A state with an unusually high lapse rate can be reviewed with its regional team. An agent with high premium per policy can be compared with peers after considering territory and product mix. A policy type with high modeled lifetime premium can receive closer service and retention attention.

Interpretation note

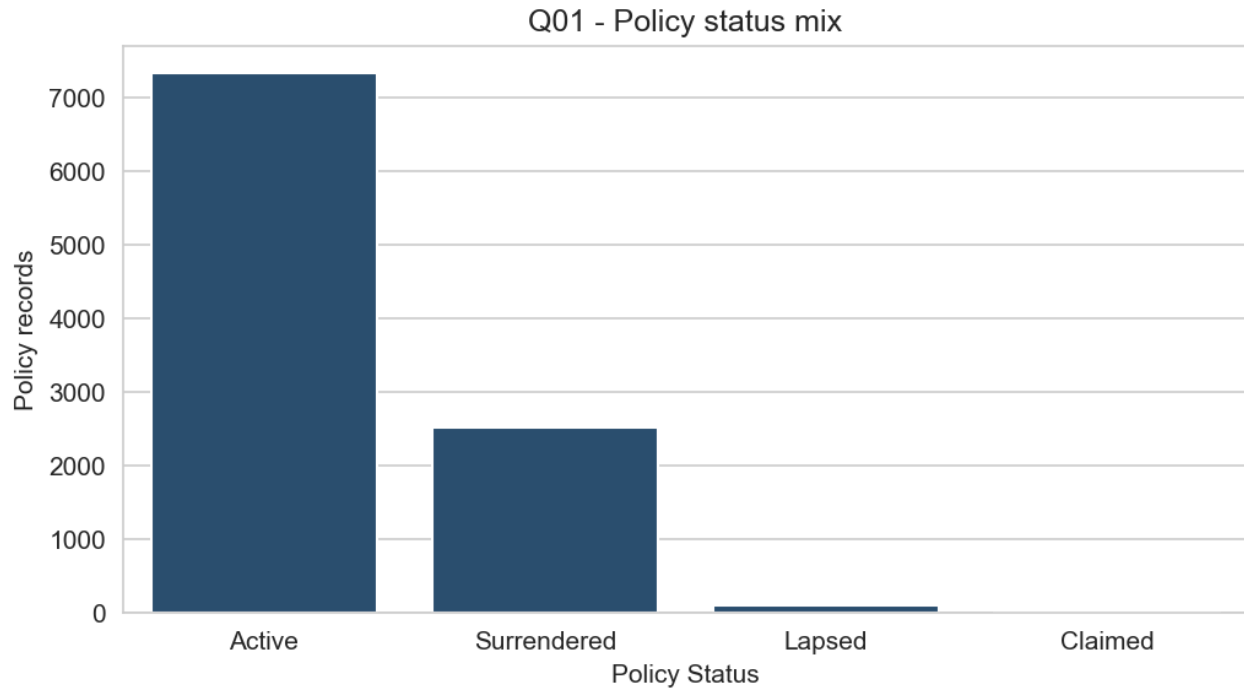
These queries show patterns, not causes. Small groups need careful interpretation, and claims, premium, and region results should be checked against approved business definitions. The source snapshot contains repeated policy numbers and incomplete regional mapping, so row counts and distinct-policy counts should remain separate.

Portfolio status mix

Q01 What does the policy status mix look like?

At a glance

7,332 active records (73.32%)



What the chart shows: Active policies dominate the stored portfolio; surrendered records are the main non-active group.

What this means: Use the status mix as a starting point for retention and servicing conversations, not as a customer-retention rate.

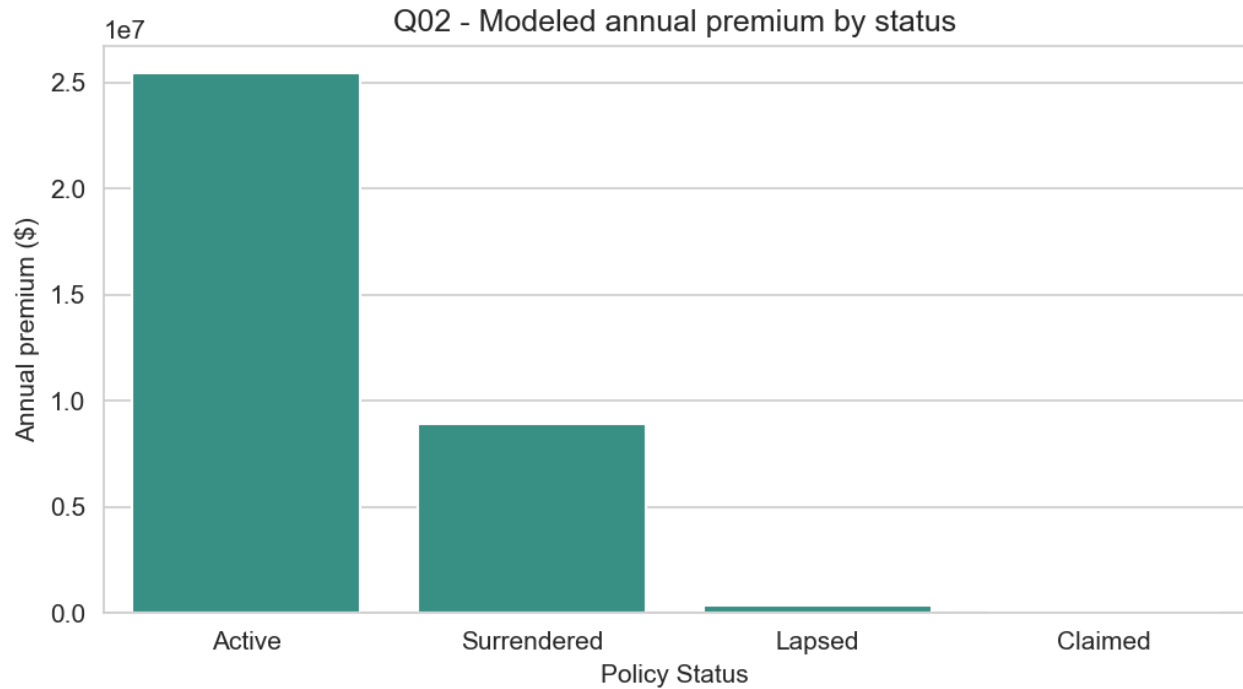
Based on the insurance portfolio source data and the stored dashboard snapshot.

Annual premium by status

Q02 Where does modeled annual premium sit by status?

At a glance

Active records hold \$25,430,623 in modeled annual premium



What the chart shows: Active records hold the largest share of modeled annual premium.

What this means: Review active premium exposure alongside lapse and surrender activity so management sees both value and risk.

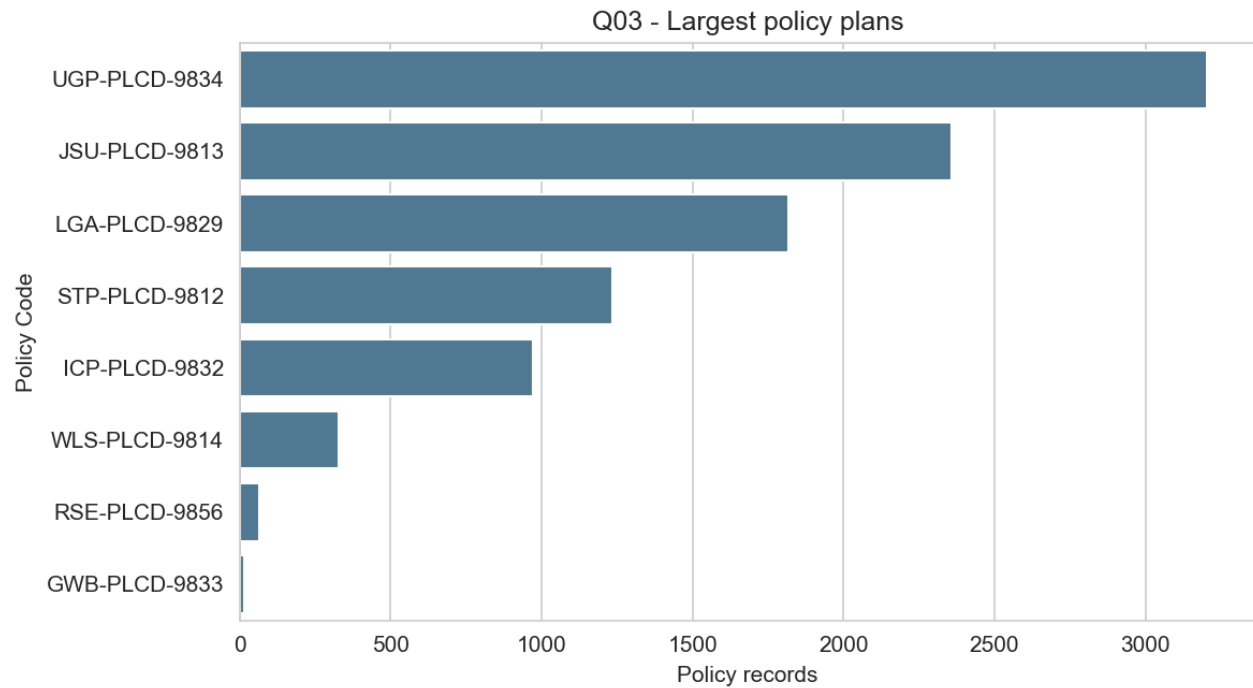
Based on the insurance portfolio source data and the stored dashboard snapshot.

Policy plan mix

Q03 Which protection plan is most common?

At a glance

UGP-PLCD-9834 is the largest plan with 3,205 records



What the chart shows: UGP-PLCD-9834 is the largest plan with 3,205 records.

What this means: The largest plans should receive priority when reviewing customer service, documentation, and data quality.

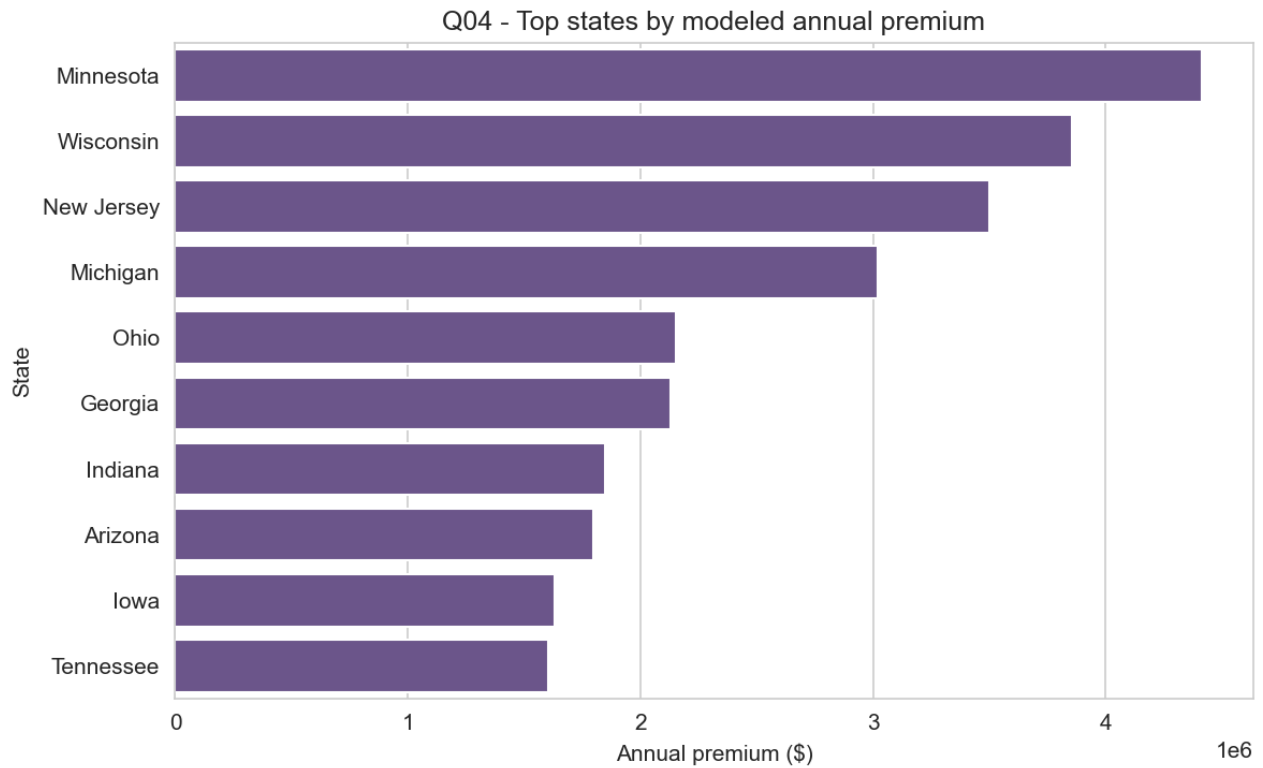
Based on the insurance portfolio source data and the stored dashboard snapshot.

Geographic premium concentration

Q04 How concentrated is annual premium by state?

At a glance

Top three states represent 33.77% of modeled annual premium



What the chart shows: The top three states represent 33.77% of modeled annual premium.

What this means: State mapping and regional reporting need to be reliable because a small number of states drive a meaningful share of the portfolio.

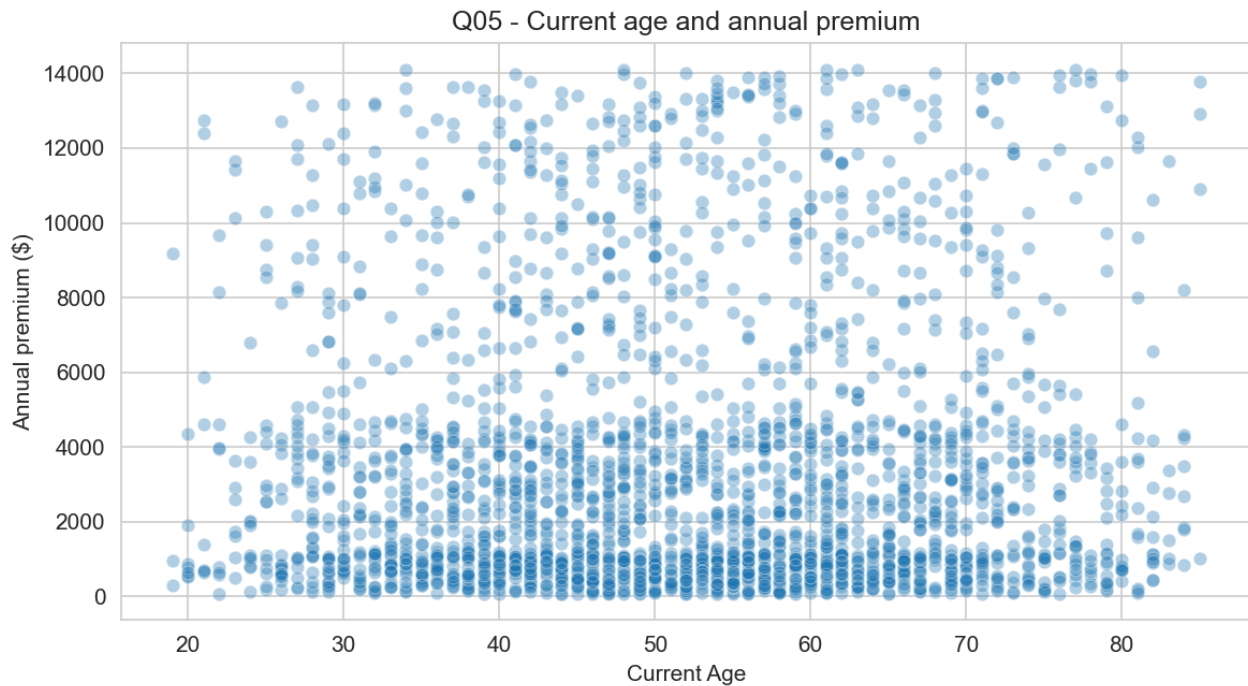
Based on the insurance portfolio source data and the stored dashboard snapshot.

Age and annual premium

Q05 Is current age related to annual premium?

At a glance

Correlation between current age and annual premium is 0.00



What the chart shows: The Pearson correlation is 0.00, showing almost no linear relationship in this snapshot.

What this means: Age alone does not explain premium size here; product, coverage, payment frequency, and underwriting factors should also be considered.

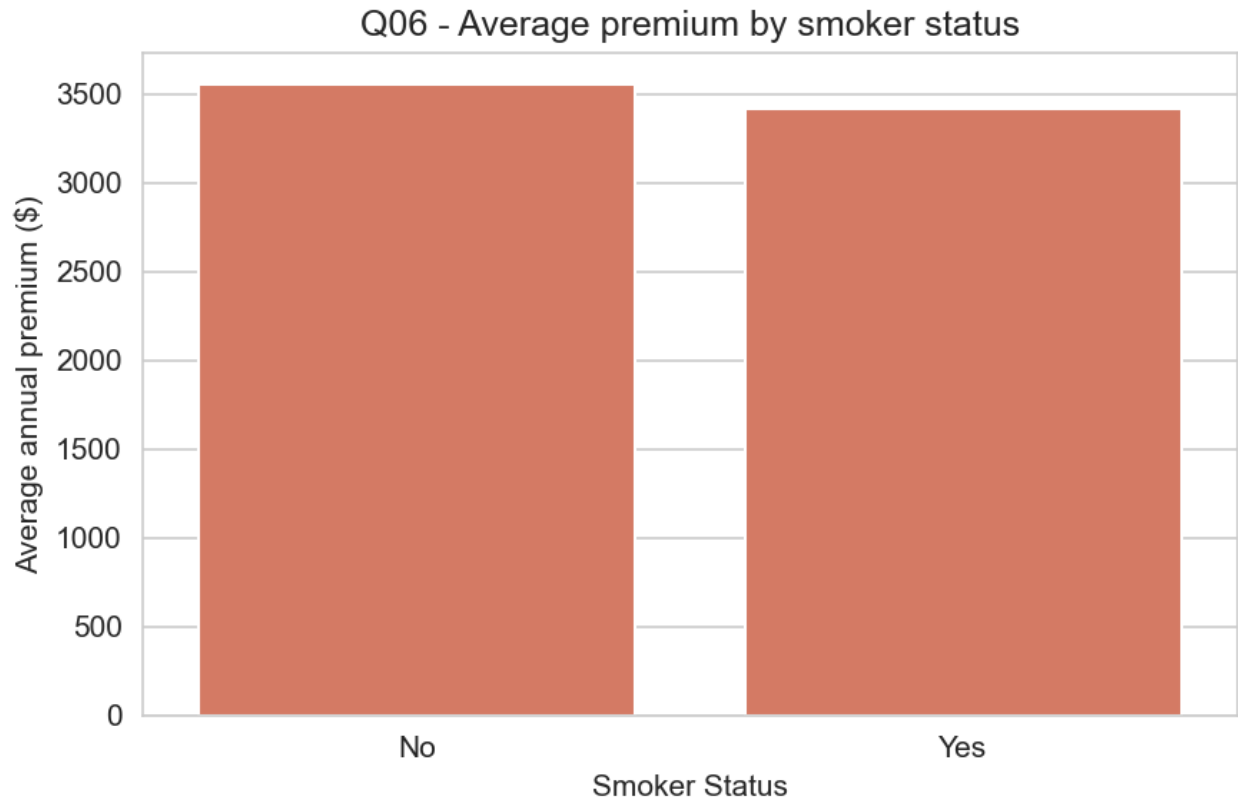
Based on the insurance portfolio source data and the stored dashboard snapshot.

Smoker status and premium

Q06 Do smoker groups have different average premiums?

At a glance

Average annual premium differs by \$136 between smoker groups



What the chart shows: Average annual premium differs by about \$136 between smoker groups.

What this means: The difference is a useful underwriting review signal, but it should be interpreted with age, coverage, and plan mix.

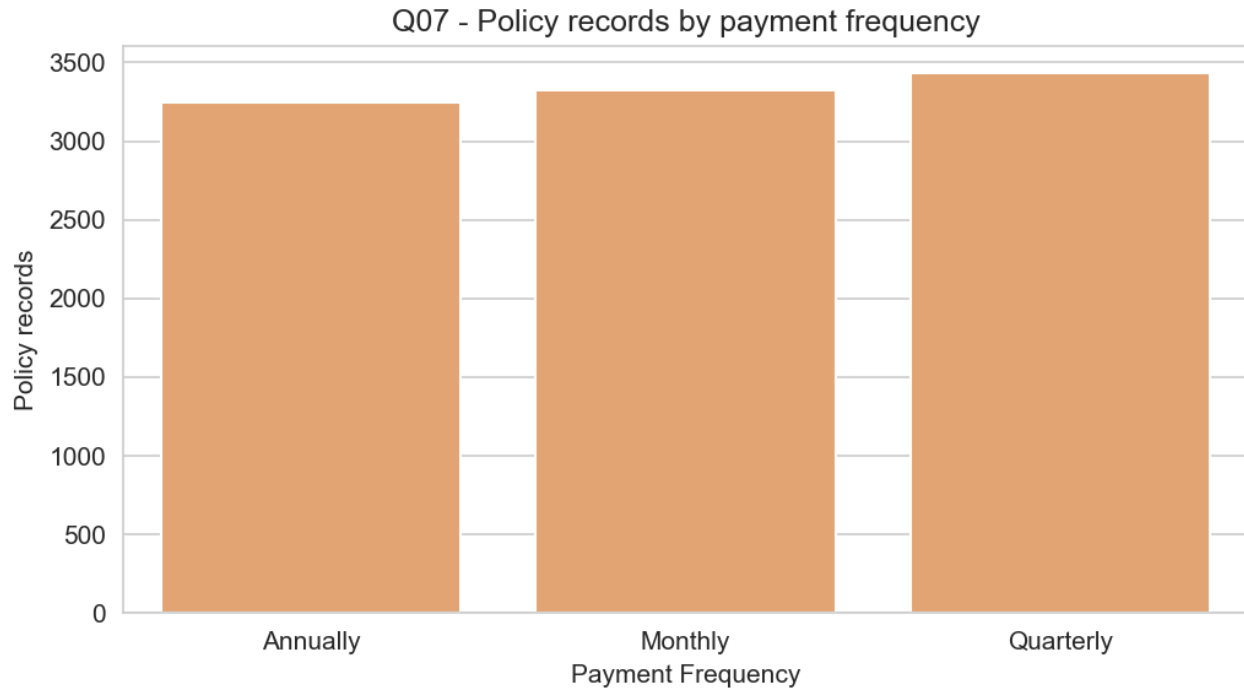
Based on the insurance portfolio source data and the stored dashboard snapshot.

Payment frequency

Q07 Which payment frequency is most common?

At a glance

Quarterly is most common with 3,432 records



What the chart shows: Quarterly payment is most common with 3,432 records.

What this means: Payment frequency affects customer cash flow and the annualization of premium, so it should remain visible in premium reporting.

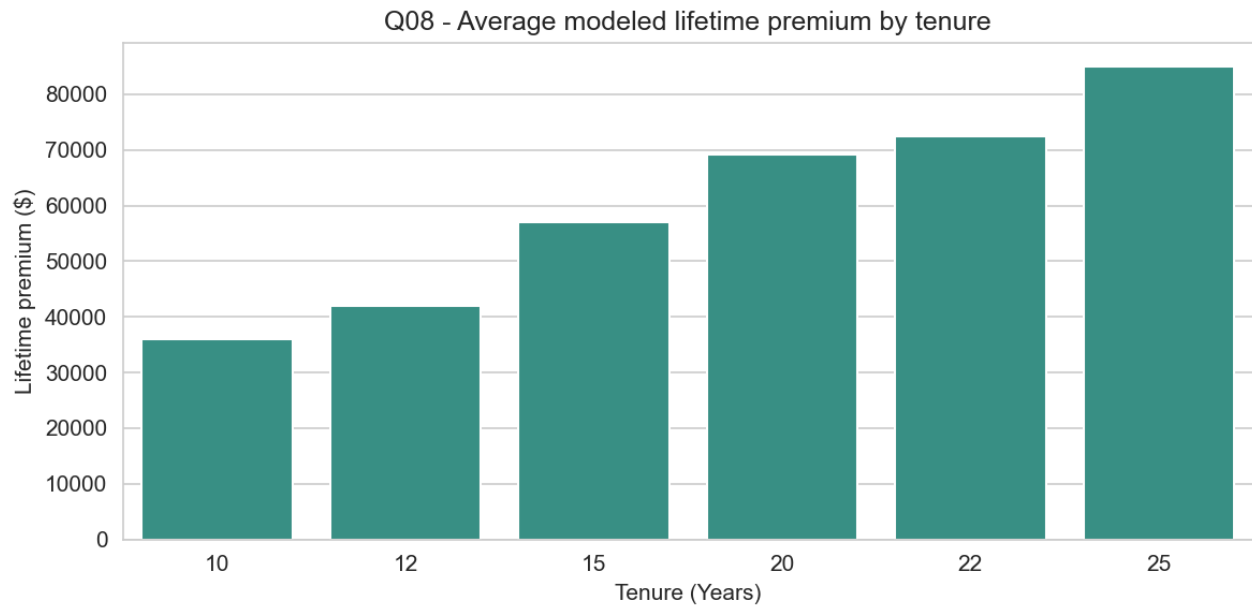
Based on the insurance portfolio source data and the stored dashboard snapshot.

Tenure and lifetime premium

Q08 Does longer tenure create more modeled lifetime premium?

At a glance

Correlation between tenure and lifetime premium is 0.25



What the chart shows: The Pearson correlation is 0.25, a modest positive relationship.

What this means: Longer tenure tends to increase modeled lifetime value, but annual premium differences are also important.

Based on the insurance portfolio source data and the stored dashboard snapshot.

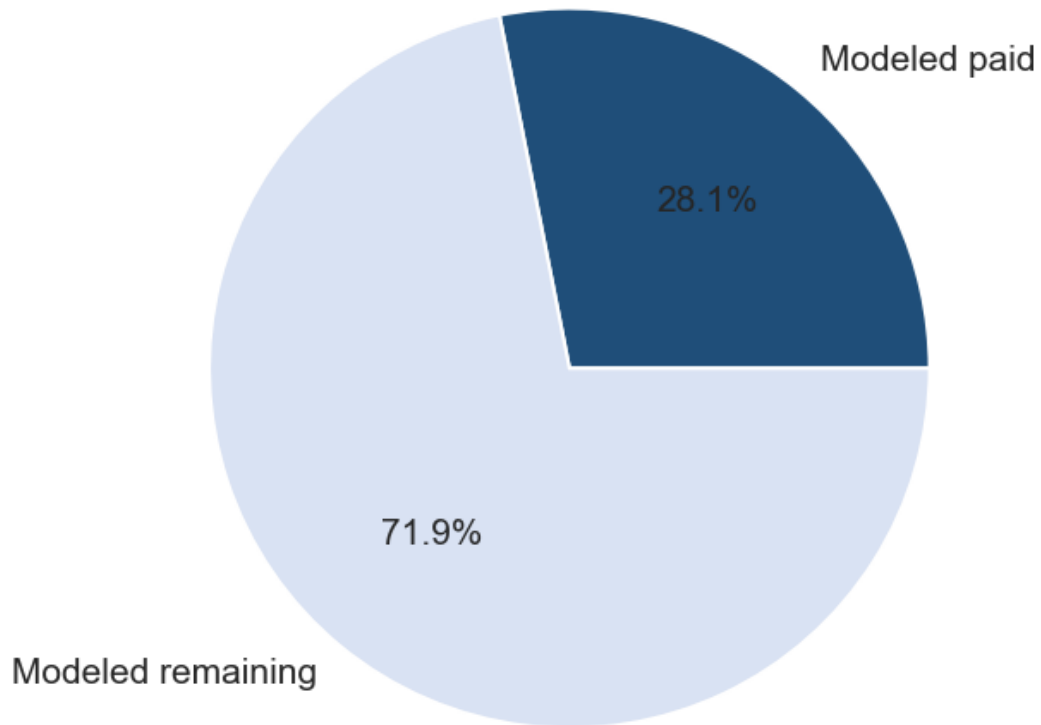
Modeled paid versus lifetime premium

Q09 How much of active lifetime premium is modeled as paid?

At a glance

Modeled paid premium is 28.12% of active lifetime premium

Q09 - Active premium composition



What the chart shows: Modeled paid premium is 28.12% of active lifetime premium.

What this means: Most modeled value remains future premium. It should not be described as cash already collected or as overdue money.

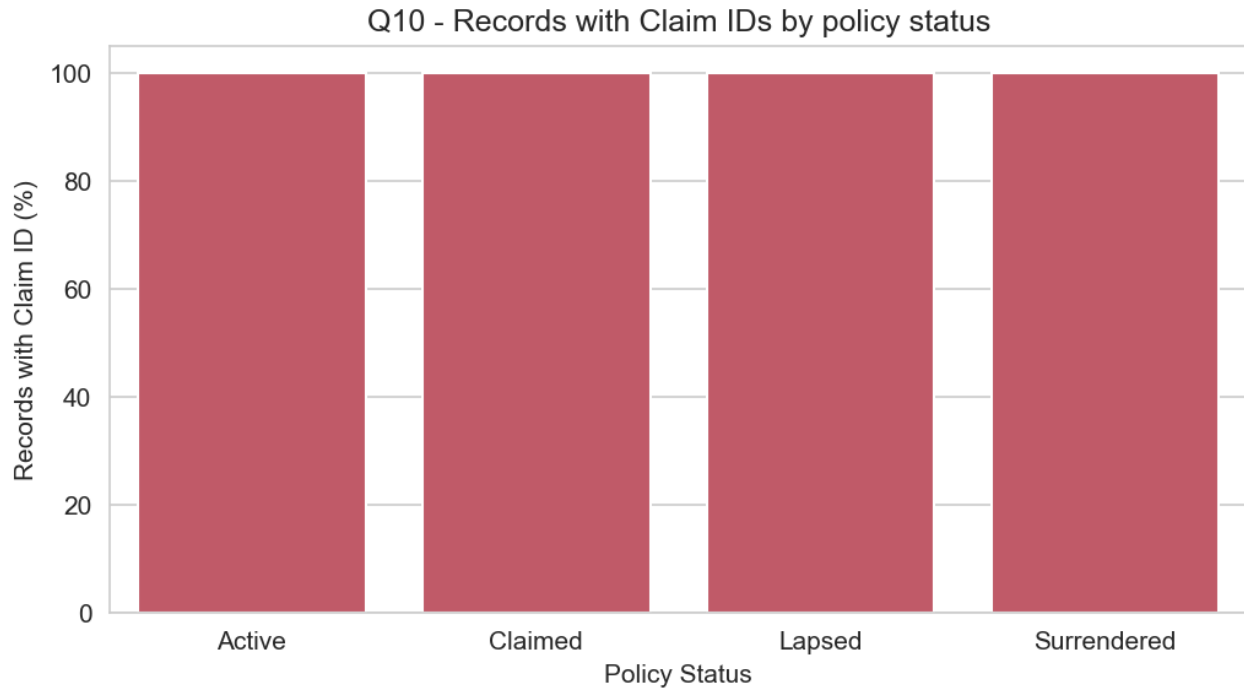
Based on the insurance portfolio source data and the stored dashboard snapshot.

Claim ID coverage by status

Q10 How are Claim IDs distributed across statuses?

At a glance

Active has the highest share of records with Claim IDs (100.00%)



What the chart shows: The source data shows Claim IDs on all Active records in this snapshot.

What this means: This is a data-definition warning. Claim ID presence should be reconciled with the business meaning of policy status before claims reporting.

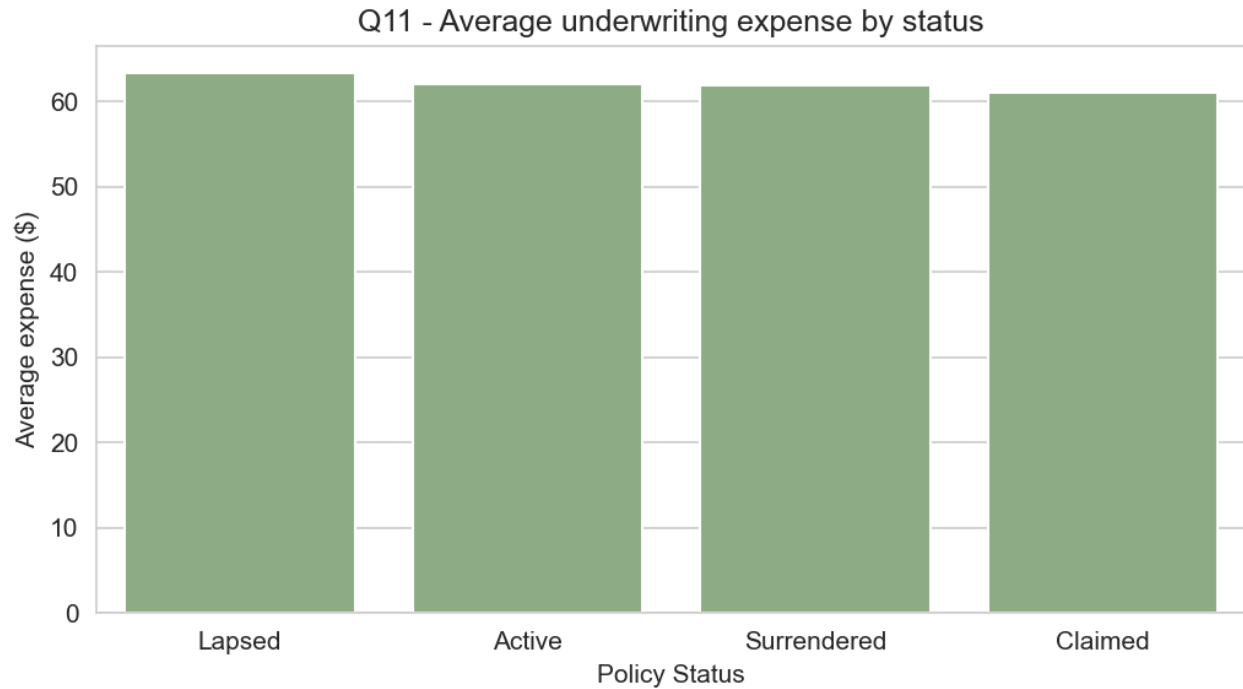
Based on the insurance portfolio source data and the stored dashboard snapshot.

Underwriting expense

Q11 Which status has the highest average underwriting expense?

At a glance

Lapsed records have the highest average expense at \$63.29



What the chart shows: Lapsed records have the highest average underwriting expense at \$63.29.

What this means: Review acquisition and servicing cost for lapsed policies to understand whether value is being lost after the initial underwriting work.

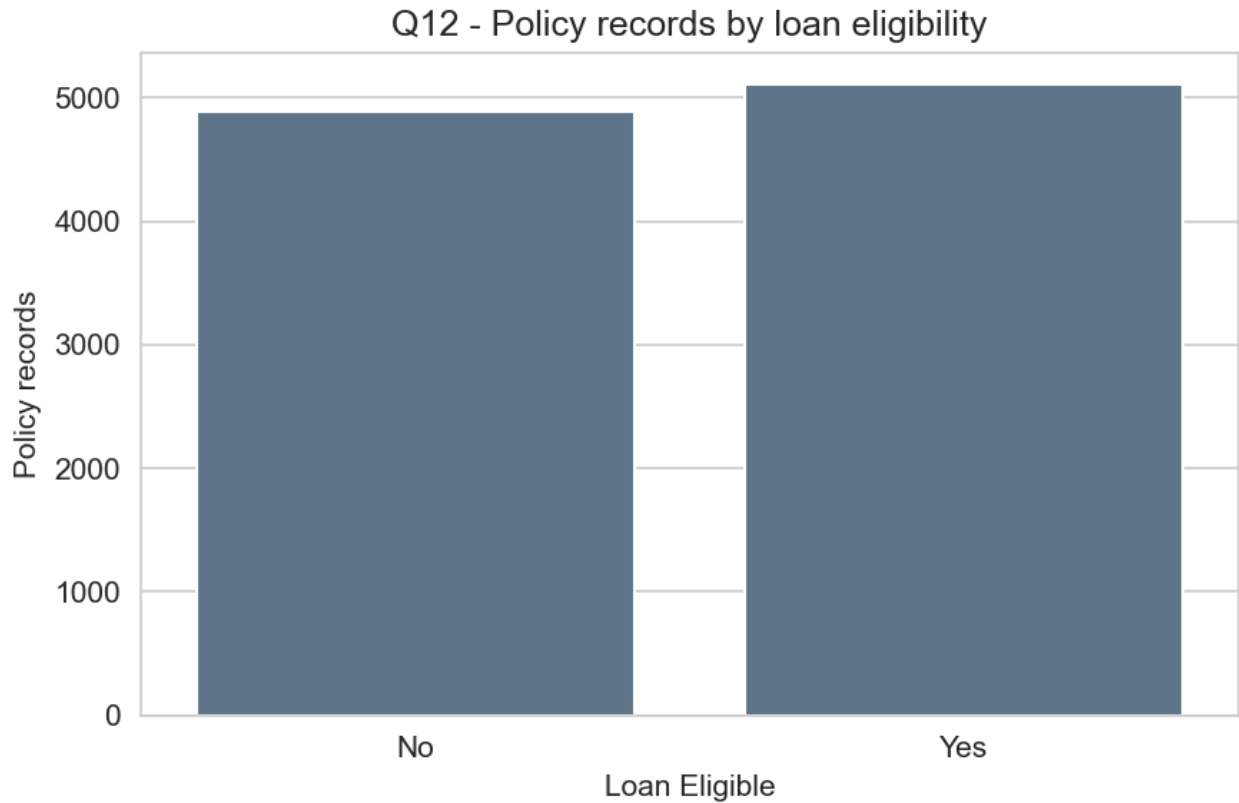
Based on the insurance portfolio source data and the stored dashboard snapshot.

Loan eligibility

Q12 How common is loan eligibility?

At a glance

51.08% of records are marked loan eligible



What the chart shows: 51.08% of records are marked loan eligible.

What this means: Loan eligibility is a useful way to compare coverage and premium segments, especially when discussing customer financial needs.

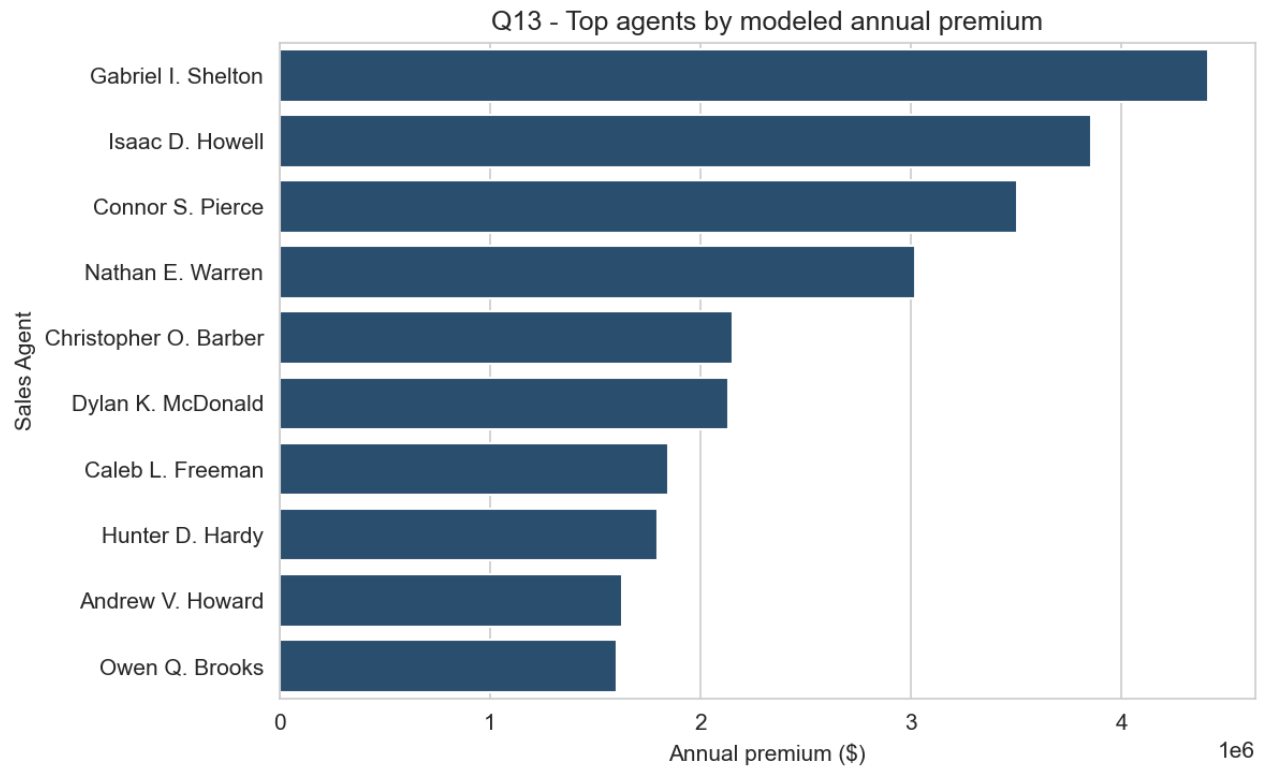
Based on the insurance portfolio source data and the stored dashboard snapshot.

Sales agent performance

Q13 Which agent manages the most modeled annual premium?

At a glance

Gabriel I. Shelton leads with \$4,412,305 modeled annual premium



What the chart shows: Gabriel I. Shelton leads with \$4.41M in modeled annual premium.

What this means: The ranking identifies portfolio responsibility; it should not be treated as a performance score without territory and mix context.

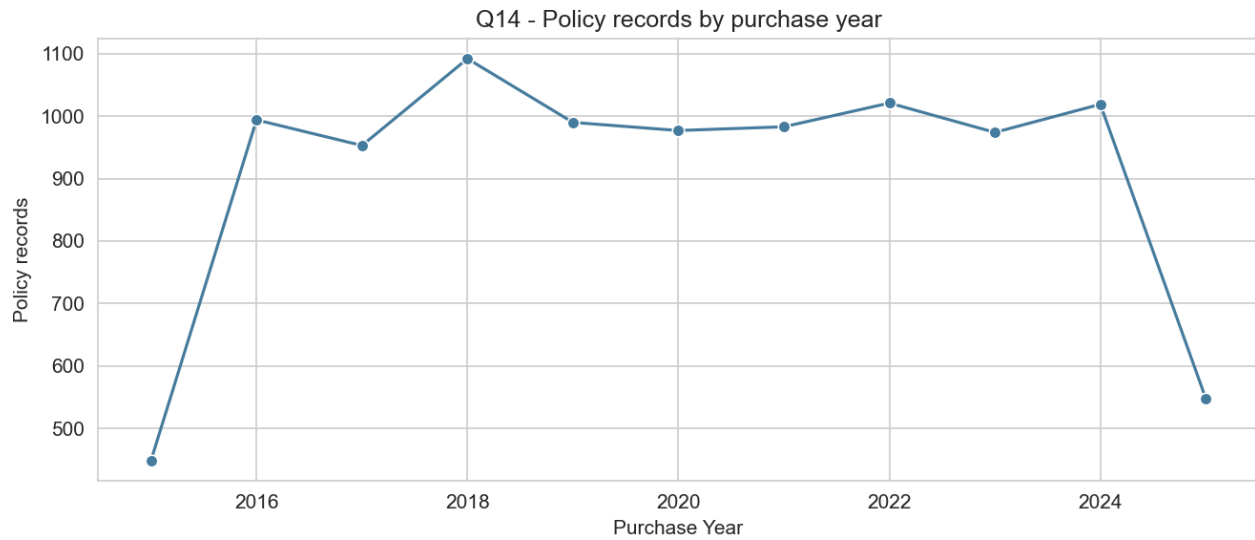
Based on the insurance portfolio source data and the stored dashboard snapshot.

Purchase-year trend

Q14 When was the largest group of policies purchased?

At a glance

2018 had the most records with 1,092



What the chart shows: 2018 had the most records with 1,092 policies.

What this means: The purchase-year pattern provides useful context for maturity, paid-premium calculations, and cohort reviews.

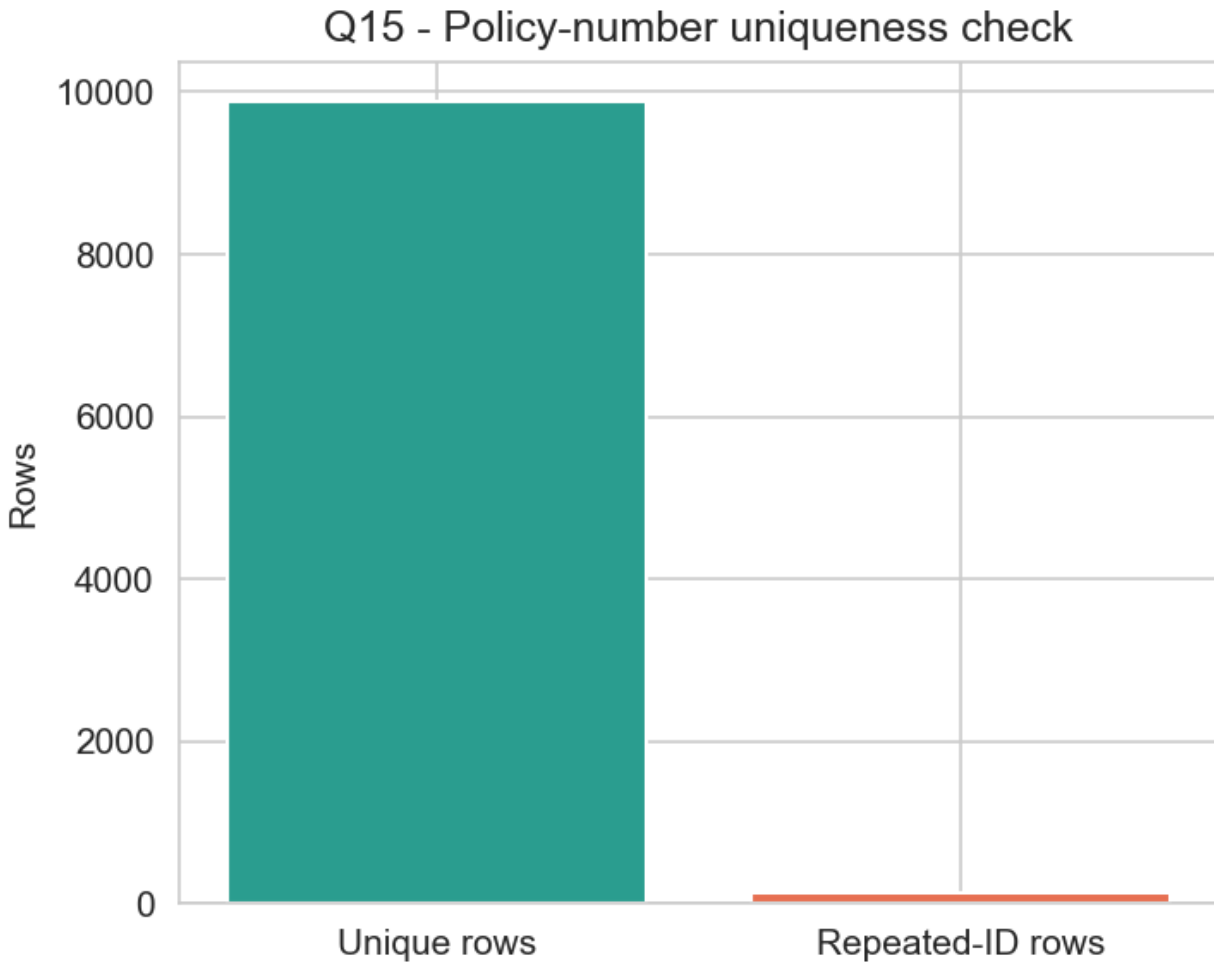
Based on the insurance portfolio source data and the stored dashboard snapshot.

Policy-number data quality

Q15 Are policy numbers unique?

At a glance

124 rows share repeated policy numbers; uniqueness is 99.38%



What the chart shows: 124 rows share repeated policy numbers; uniqueness is 99.38%.

What this means: Repeated identifiers can change policy counts and premium totals, so the reporting definition should distinguish rows from unique policies.

Based on the insurance portfolio source data and the stored dashboard snapshot.